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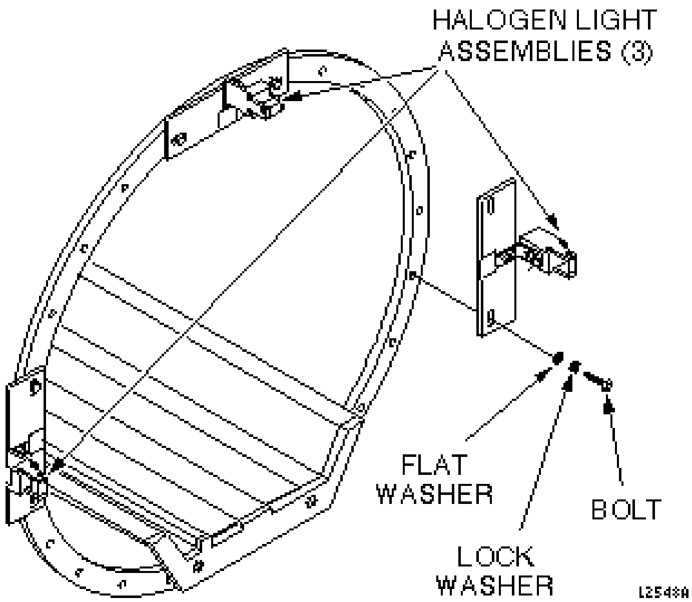
Description - Procedures for aligning the patient alignment lights and checking the power supply voltage. For CX systems refer to procedure for Patient Alignment Lights for CX Magnet.

1- HALOGEN LIGHT ALIGNMENT PROCEDURES

The Halogen Light Assemblies are mounted on the front magnet frame as shown in Illustration 1-1. All three assemblies are identical.



HALOGEN LIGHT ASSEMBLY LOCATIONS
ILLUSTRATION 1-1

The light assemblies are preadjusted at the factory and, after installation in the field, require that the procedures in step 11 be followed to ensure that the halogen beams are in proper alignment. After the initial set up and alignment, the assemblies do not normally require adjustment or maintenance other than bulb replacement. However, in the event that the assemblies become misaligned, the following alignment procedure must be followed in the sequence specified. Before performing this procedure, observe the DANGER sign on the upper light assembly bracket, and observe the precautions noted in the following warning statement:

WARNING!

POSSIBLE PERSONAL INJURY! THE HALOGEN BULB CAN EASILY SHATTER IF SCRATCHED OR DAMAGED. ALSO, THE BULB AND ITS FIXTURE GET EXTREMELY HOT DURING USE. WEAR PROTECTIVE EYEWEAR AND LIGHT NONABRASIVE GLOVES. REMOVE GREASE AND FINGER PRINTS FROM BULB BY CLEANING WITH A GREASE-FREE SOLVENT. ENSURE THAT POWER IS NOT APPLIED TO BULBS DURING CLEANING. ENSURE THAT EACH HALOGEN BULB IS FIRMLY INSTALLED IN SOCKET.

Tools Required

- SPT Short Loader Cylinder, 2135652

Note

The SPT Short Loader Cylinder is part of the SPT Short Loader Assembly (2125244). This hardware, though shipped with Signa systems, is the property of GE. It is available for customer purchase.

- Nonmagnetic scale (48 in. nominal)
- Nonmagnetic 12 in. ruler
- Nonmagnetic screwdriver
- Nonmagnetic 0.050 in. Allen wrench
- Nonmagnetic open-end wrench
- Multimeter
- Protective eyewear
- Nonabrasive gloves
- Nonmagnetic straight edge (min. 43 in.; max. 45 in.)

WARNING!

POISON HAZARD! THE PHANTOM CONTAINS NICKEL, A SUSPECTED CARCINOGEN. DO NOT INGEST. DISPOSE OF AS A HAZARDOUS WASTE ACCORDING TO STATE AND FEDERAL REGULATIONS.

Procedure

1. Open magnet enclosure front cover.

CAUTION

Equipment damage possibility. Do not twist halogen bulbs in the lampholder sockets. The bulb has a two-pin connector that is mounted in a lampholder socket. The bulb must be firmly pushed in to the socket to be mounted properly. It will be damaged if it is twisted or turned. Ensure that power to bulbs is off before working with them; they get extremely hot during operation.

2. Remove screws securing cover over halogen bulbs and lift cover from each Halogen Light Assembly, taking care not to damage the rear lens alignment pin that protrudes though the side of the cover. Ensure that each bulb is pushed in securely, and is perfectly straight in its socket.

Note

A misalignment of the bulb will distort the beam path.

3. Replace covers on Halogen Light Assemblies, and secure with mounting screws.

Note

All further adjustments on the Halogen Light Assemblies should be performed with the protective covers installed. This minimizes the chance of touching the bulb while performing adjustments, as well as getting any foreign materials on the bulb. The protective cover is removed only for bulb replacement.

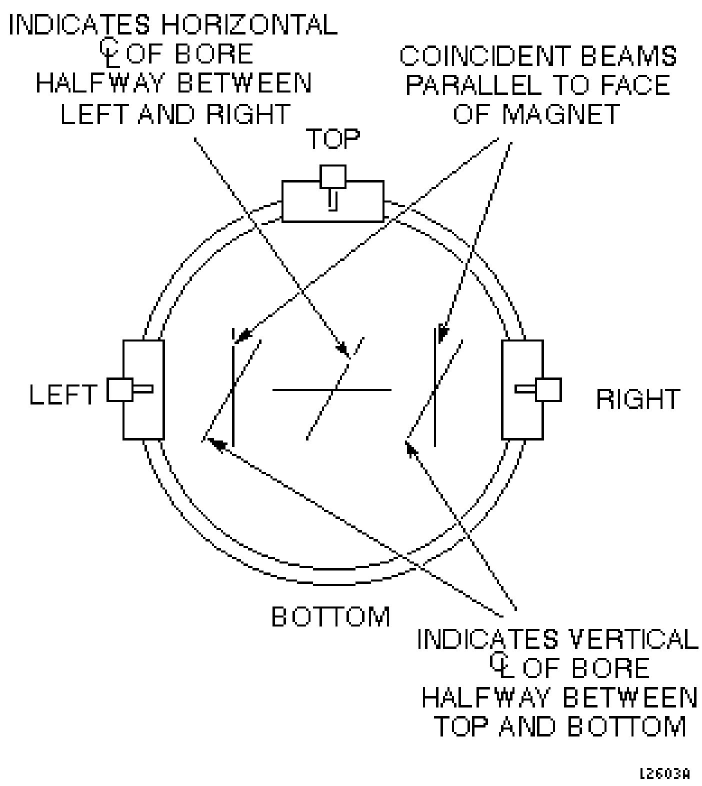
4. Turn on power to the Halogen Light Assemblies High Current Power Supply Board (MR1A14A2) by switching on CB6 on Magnet Enclosure Power Supply Module (MR1A14) at the RF/Pen Cabinet.

   For RF/Pen II and RF/PDU Cabinets, switch on circuit breaker MR1A20 CB4 on the System Support Module (SSM).

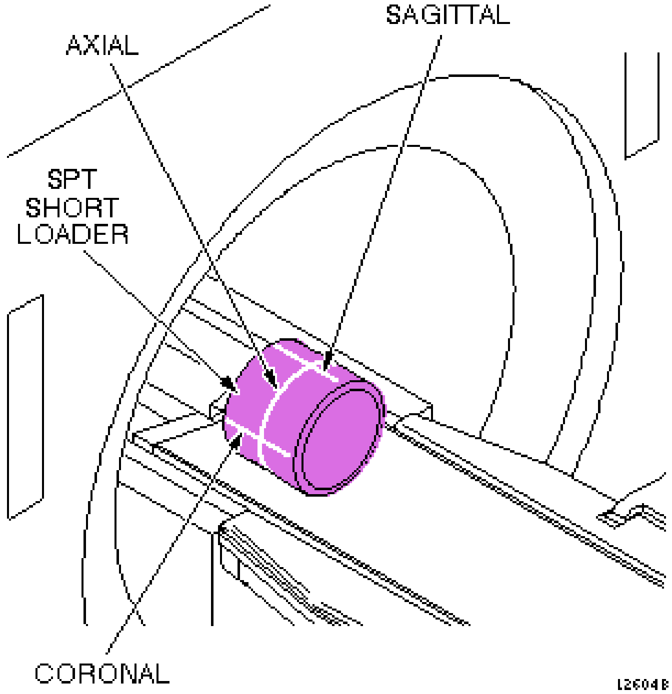
5. Place SPT Short Loader Cylinder on patient table.

6. Press ALIGN ON pushbutton on front of magnet enclosure.

7. Close magnet enclosure front cover and observe light beam at top aperture. Verify that light beam is unobstructed, and that the mirror is centered in the aperture, and that there is no interference with the magnet cover. See Illustration 1-2 and Illustration 1-3 for the desired beam pattern at the front of the magnet enclosure.

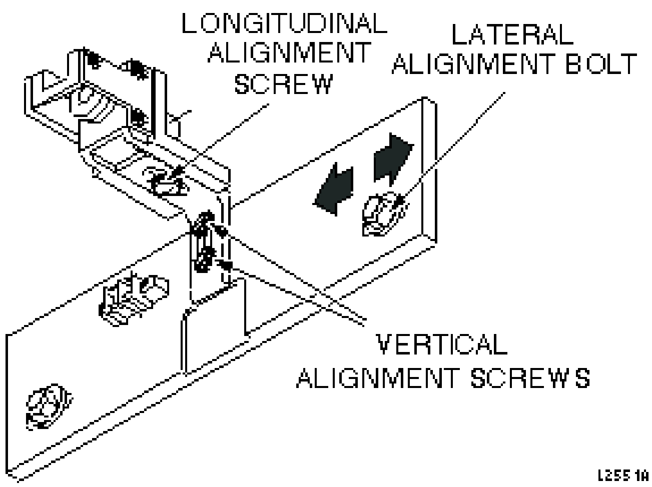


HALOGEN LIGHT BEAM PATTERN
ILLUSTRATION 1-2



HALOGEN WRAPAROUND BEAM PATTERN
ILLUSTRATION 1-3

8. If mirror is not centered in the aperture, or if there is an interference with the front cover, open front magnet enclosure cover and adjust the Halogen Light Assemblies by loosening the vertical and/or longitudinal screws (see Illustration 1-4).



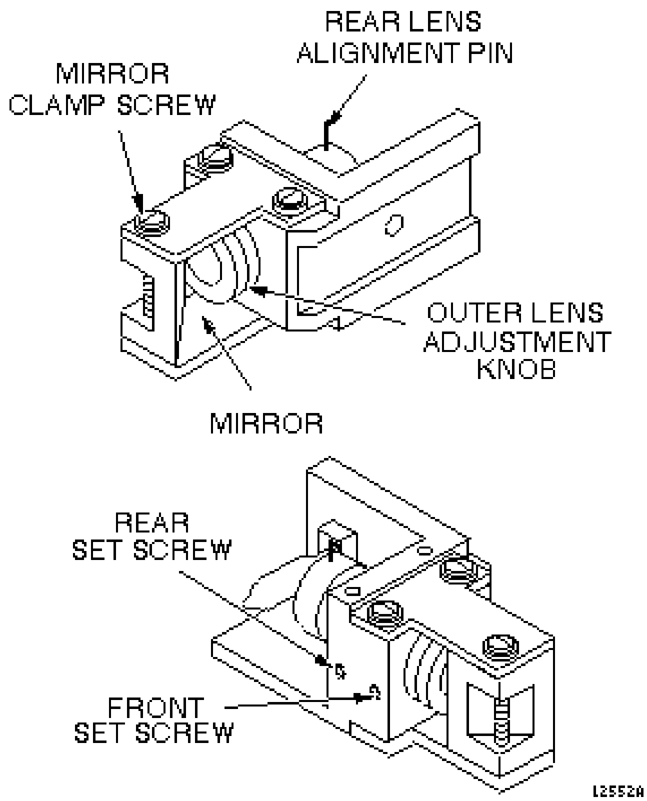
UPPER HALOGEN LIGHT ASSEMBLY
ILLUSTRATION 1-4

9. Adjust the three assemblies in the lateral direction so that the beams are centered in the magnet bore as shown in Illustration 1-2 and Illustration 1-3 above. Illustration 1-3 shows the wraparound pattern with the three halogen beams perfectly aligned. The top halogen sagittal beam should intersect the bore at the horizontal centerline. This can be measured by placing a nonmagnetic scale horizontally across the face of the bore and measuring the intersection of the beam.

The left and right halogen coronal beams should intersect the bore at the vertical centerline of the bore opening. A non-magnetic scale placed vertically across the face of the bore will determine the alignment of these beams to the vertical centerline of the bore.

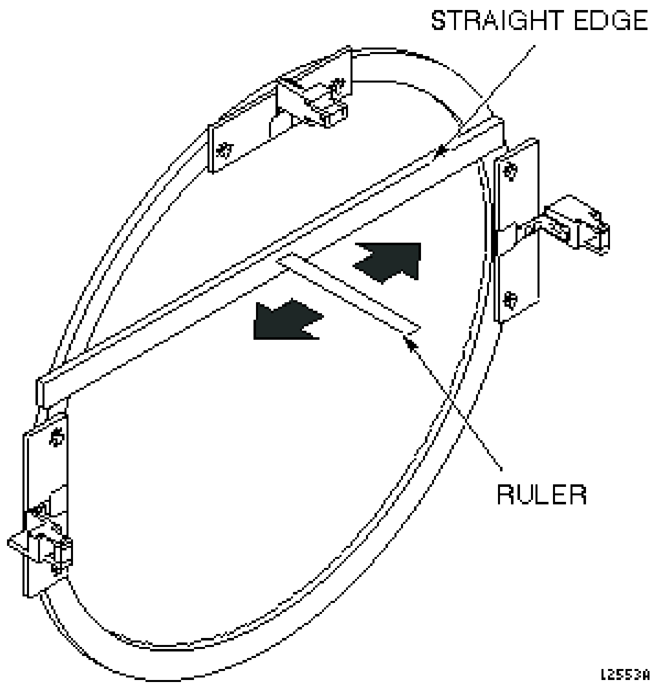
10. Tighten alignment screws when adjustment is complete.

11. If beams are out of focus, use a 0.050-inch Allen wrench and loosen front set screw on top of the top Halogen Light Assembly (see Illustration 1-5). Turn outer lens adjustment until image is sharply in focus. Use a piece of cardboard or paper on front edge of bridge to determine image focus quality. Perform the focus adjustment on each of the Halogen Light Assemblies in turn. Tighten set screw on each assembly when focus adjustment is complete.



HALOGEN LIGHT ASSEMBLY ADJUSTMENTS
ILLUSTRATION 1-5

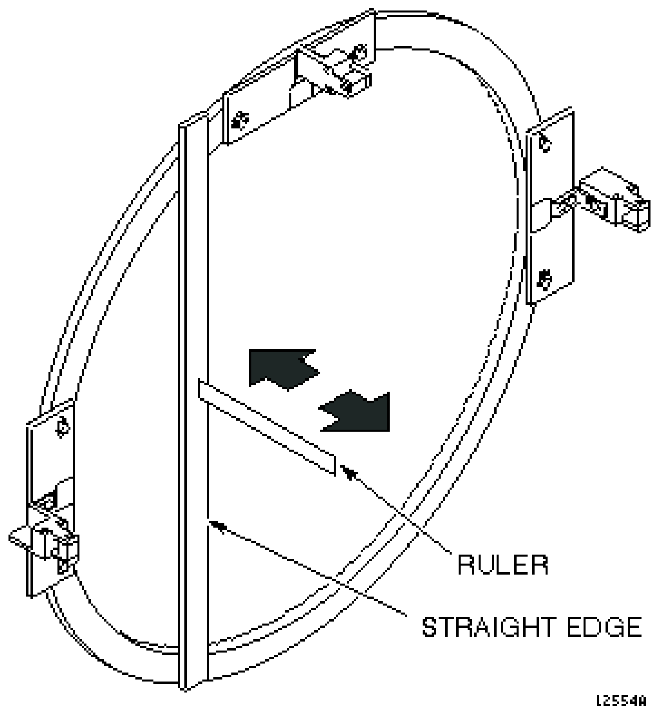
12. Using a nonmagnetic straight edge, measure axial beam squareness to face of magnet as follows:
    a. Place straight edge across the face of the magnet frame, as shown in Illustration 1-6.



UPPER HALOGEN ASSEMBLY ALIGNMENT
ILLUSTRATION 1-6

    b. Place a nonmagnetic ruler against the straight edge, and check the squareness of the top halogen axial beam by sliding the ruler along the length of the straight edge. Take several readings to ensure that axial beam strikes ruler at exact location at each measurement point; this indicates that beam is square to face of magnet.
    c. If beam alignment is off, loosen rear set screw and gently move rear lens alignment pin until beam is aligned. (See Illustration 1-5. above)
    d. Tighten rear set screw.

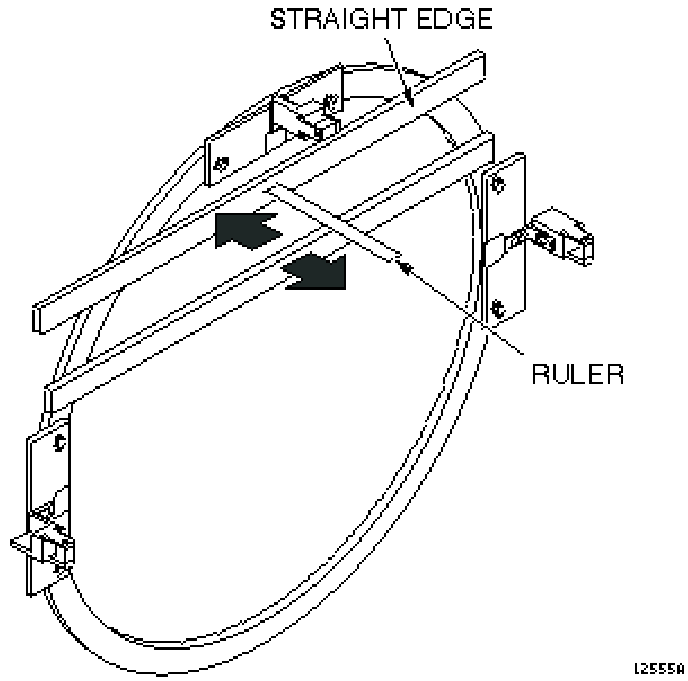
e. Repeat this procedure for left and right Halogen Light Assemblies, by placing the straight edge against the magnet face as shown in Illustration 1-7. Hold straight edge in one hand and, in the other hand, slide a ruler up and down the face of the straight edge, observing that the axial beam strikes the ruler in the exact location at each measurement point, to ensure that beam is square to face of magnet.



L/R HALOGEN ASSEMBLY ALIGNMENT
ILLUSTRATION 1-7

f. If measurement points are off, loosen rear set screw and gently move rear lens alignment pin until beam is aligned.
g. Tighten rear set screw on left and right Halogen Light Assemblies.

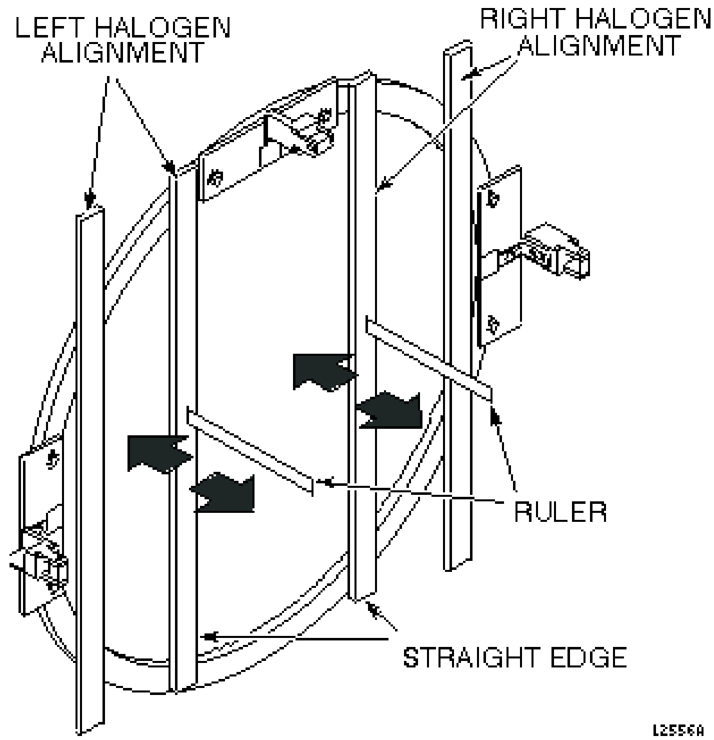
h. Measure that the axial beams are parallel to face of magnet by loosening mirror clamping screw on upper Halogen Light Assembly (see Illustration 1-5 above). Place straight edge across face of magnet in the positions shown in Illustration 1-8, and tilt mirror face to ensure that the axial beam strikes the ruler in the exact location.



UPPER HALOGEN MIRROR ANGLE BEAM
ILLUSTRATION 1-8

i. Tighten mirror clamping screw.

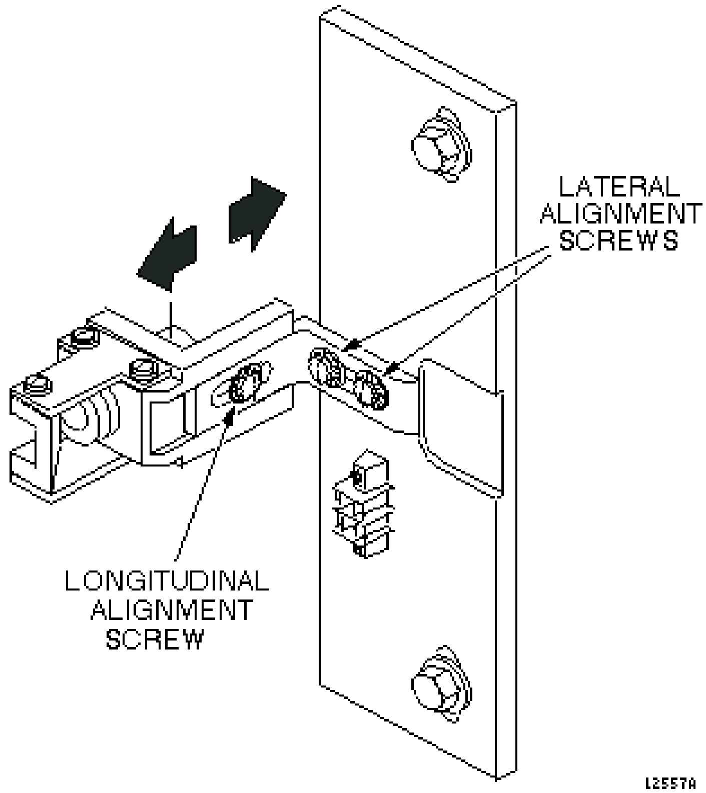
j. Repeat step *h*. for the left and right Halogen Light Assemblies (see Illustration 1-9).



L/R HALOGEN MIRROR ANGLE BEAM
ILLUSTRATION 1-9

k. Tighten mirror clamping screw on each of the Halogen Light Assemblies when alignment procedure is completed.

l. Perform final alignment by loosening longitudinal alignment screws on bracket of left and right Halogen Light Assemblies to allow free in/out movement (see Illustration 1-10). Align left and right axial beams with the top axial beam. Adjust assemblies to ensure that beam path is normal (see Illustration 1-2 and Illustration 1-3 above).



L/R HALOGEN ADJUSTMENT (LEFT SHOWN)
ILLUSTRATION 1-10

m. Tighten bracket screws when beam paths are normal.

13. Perform a final check of beam pattern, as shown in Illustration 1-2 and Illustration 1-3 above. Further adjustments may be required to bring beam patterns into exact alignment. Normally, fine tuning of the rear lens and the mirror accomplish this.

14. Remove all tools and close front magnet enclosure cover.

2- PATIENT ALIGNMENT LIGHTS POWER SUPPLY

2-1 +12.75 Vdc Supply

The +12.75 Vdc Patient Alignment Lights (PAL) power supply, which supplies power to the Halogen Light Assemblies, does not normally require calibration or maintenance. However, if this power supply is field replaced, its voltage output should be calibrated. The PAL power supply is located in the Magnet Enclosure Power Supply Module (MR1A14), on the High Current Power Supply Board (MR1 A14A2) located on the lid. Refer to procedure for Magnet Enclosure P.S. Module Calibration.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 16, 1998	J. Saperstein	Initial conversion from Toolbook to Word.
1	January 6, 1999	K. Keshena	Updates to styles.